# THE UNIVERSITY OF DODOMA

## GROUP ASSIGNMENT

GROUP NUMBER : 09

MODULE NAME : MICROCONTROLLER

MODULE CODE : CT 321

FACILITATOR : Dr. CRALLET VICTOR

SEMESTER : TWO

ACADEMIC YEAR : 2025 / 2026

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# Contents

Table of Figures

# 1. Introduction and Project Background

## 1.1 Context and Overview

In modern automated manufacturing, processing facilities, and logistics centers, the ability to accurately detect, track, and catalog moving components along a conveyor or production line is a fundamental requirement of industrial automation. Automated counting systems provide critical data points for inventory tracking, throughput optimization, safety protocols, and quality assurance workflows. At the core of these automated pipelines are embedded microcontrollers designed to process high-frequency, non-periodic input pulses generated by optical, capacitive, or mechanical sensors as physical items break a sensory field.

This project centers on the design, architectural formulation, and structural implementation of a reliable **External Event Object Counter** built around the high-performance, low-power **8-bit ATmega32 AVR RISC-based microcontroller**. The system acts as a localized edge device that captures incoming asynchronous pulses, updates internal counting registries, and transmits alphanumeric telemetry values across an isolated parallel data bus to a 16x2 Liquid Crystal Display (LCD) screen, while concurrently driving a physical status beacon LED.

## 1.2 Problem Statement and Engineering Constraints

The fundamental engineering challenge when designing real-time monitoring instruments lies in the management of system latency and processing bottlenecks. A traditional, naive firmware approach relies on continuous loop polling—where the central processing unit (CPU) repeatedly executes conditional check statements (such as reading the digital logic state of an input pin) inside a main while(1) program loop.

While polling is straightforward to implement, it introduces severe architectural flaws when scaling up to high-speed or multi-peripheral systems:

- **Missed Triggers:** If an object passes a sensor field very rapidly, generating a narrow pulse-width on the order of milliseconds, the event can pass entirely undetected if the CPU happens to be executing alternate code (such as rendering characters or executing time delays for an LCD display) during that exact window.
- **CPU Inefficiency:** Polling forces the microcontroller to consume clock cycles continuously just to read an idle input line, drastically reducing the bandwidth available for running secondary calculations, managing communications protocols, or processing complex control signals.

To bypass these operational limitations, this design replaces polling routines with a hardware-driven **External Interrupt Architecture ($INT0$)** bound to edge-detection logic.

## 1.3 System Design Objectives

The operational and structural execution of this engineering project is governed by four primary design directives:

1. **Low-Latency Interrupt Processing:** Program internal hardware registers to flag execution requests strictly on the **falling edge** of incoming sensor signals, redirecting the program pointer immediately to an optimized Interrupt Service Routine (ISR) to secure 100% pulse acquisition accuracy at input frequencies up to 50 Hz.
2. **Deterministic UI Updates:** Dedicate the microcontroller's main processing loop purely to formatting integer variables into ASCII strings and driving an **8-bit parallel bus across Port C** to refresh the alphanumeric display grid without interfering with input capturing.
3. **Hardware Layout Optimization:** Utilize the KiCad Electronic Design Automation (EDA) suite to map out a production-ready, highly compact **single-layer Printed Circuit Board (PCB)** layout. The engineering constraints demand a jumperless single-layer routing topology, utilizing advanced spatial layouts such as rounded, curved trace paths and central-core bridging beneath the integrated circuit (IC) socket to minimize board space.

4. **Verifiable Testing & Configuration Management:** Validate all firmware logic inside a hardware-emulated environment using **SimulIDE** to verify signal pathways under dynamic pulse counts before moving to manufacturing. Finally, organize the entire project lifecycle inside an industry-standard, clean **GitHub repository** structure using structured commit tracking to facilitate assessment, collaborative reviews, and version control management.

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## 2. System Architecture & Peripheral Interfacing

The architecture of the system is carefully split into isolated processing, signal detection, and visual display blocks to maximize data bandwidth and eliminate processing latency.

### 2.1 Complete Hardware Pin Mapping Matrix

| Peripheral Component | ATmega32 Hardware Pin / Port Reference | I/O Direction Status | Electrical & Register Configuration |
|---|---|---|---|
| **LCD Parallel Data Bus (D0–D7)** | Port C (`PC0`, `PC1`, `PC2`, `PC3`, `PC4`, `PC5`, `PC6`, `PC7`) | Output | Push-Pull, High-Drive Sourcing State (`DDRC = 0xFF`) |
| **LCD Register Select (RS)** | Dedicated GPIO Line | Output | Digital Level Control |
| **LCD Enable Clock (E)** | Dedicated GPIO Line | Output | High-to-Low Strobe Triggering Pulse |
| **Event Counter Button (Sensor)** | Port D Pin 2 (`PD2` / `INT0`) | Input | Internal Hardware Pull-Up Resistor Enabled |
| **Status Milestone LED** | Port B Pin 2 (`PB2`) | Output | Active-High Current Source (`DDRB \|= (1 << PB2)`) |

### 2.2 Circuit Simulation and Verification

Before moving to physical board layouts, the entire circuit schematic and register configurations were built and dynamically validated in a real-time simulation environment. This step verified that the hardware interrupt triggers and LCD data byte transfers executed with correct clock synchronization.

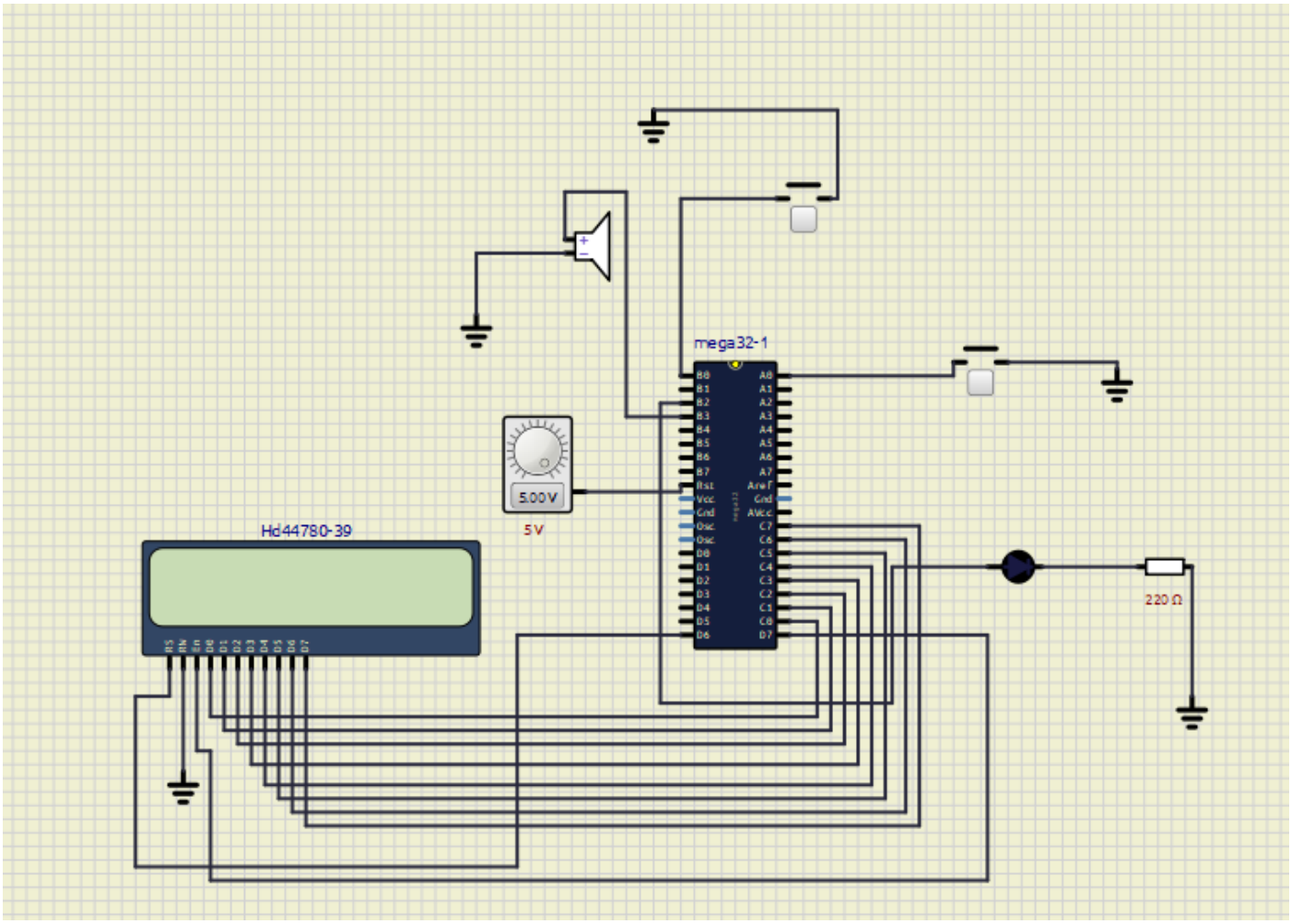



*Figure 1:Full schematic simulation in SimulIDE*

*Full schematic simulation in SimulIDE showing the ATmega32, the push-button trigger tied to INT0, and the active 16x2 LCD layout.*

## 3. Printed Circuit Board (PCB) Design & Layout Strategy

The physical board layout was compiled within the KiCad PCB Design suite. The layout emphasizes reliability, ease of manual assembly in a university laboratory environment, and elegant trace aesthetics.

### 3.1 Layout Topology & Track Configuration

- **Single-Layer Bottom Copper Constraints:** To ensure the design could be reliably etched, drilled, and hand-soldered without complex double-sided alignment or via-plating machinery, the entire circuit is routed exclusively on the Bottom Copper layer (**B.Cu**).
- **Advanced Curved Trace Architecture:** Standard sharp trace angles introduce layout bottlenecks when routing parallel buses. By configuring and deploying **Rounded Arc Tracks**, the 8 parallel data lines running from Port C to the LCD header wrap smoothly around the board contours like a uniform ribbon cable. This layout approach saves significant board surface area and guarantees neat, professional trace parallelisms.

- **Component-Level Path Bridging:** To overcome routing barriers on a single layer without adding jumpers, paths are guided directly beneath the dead space located in the center of the through-hole ATmega32 Dual In-line Package (DIP) socket. Furthermore, axial resistors (R1 to R5) are utilized strategically as physical layout bridges, allowing copper traces to pass safely through the empty gap between their mounting pads.

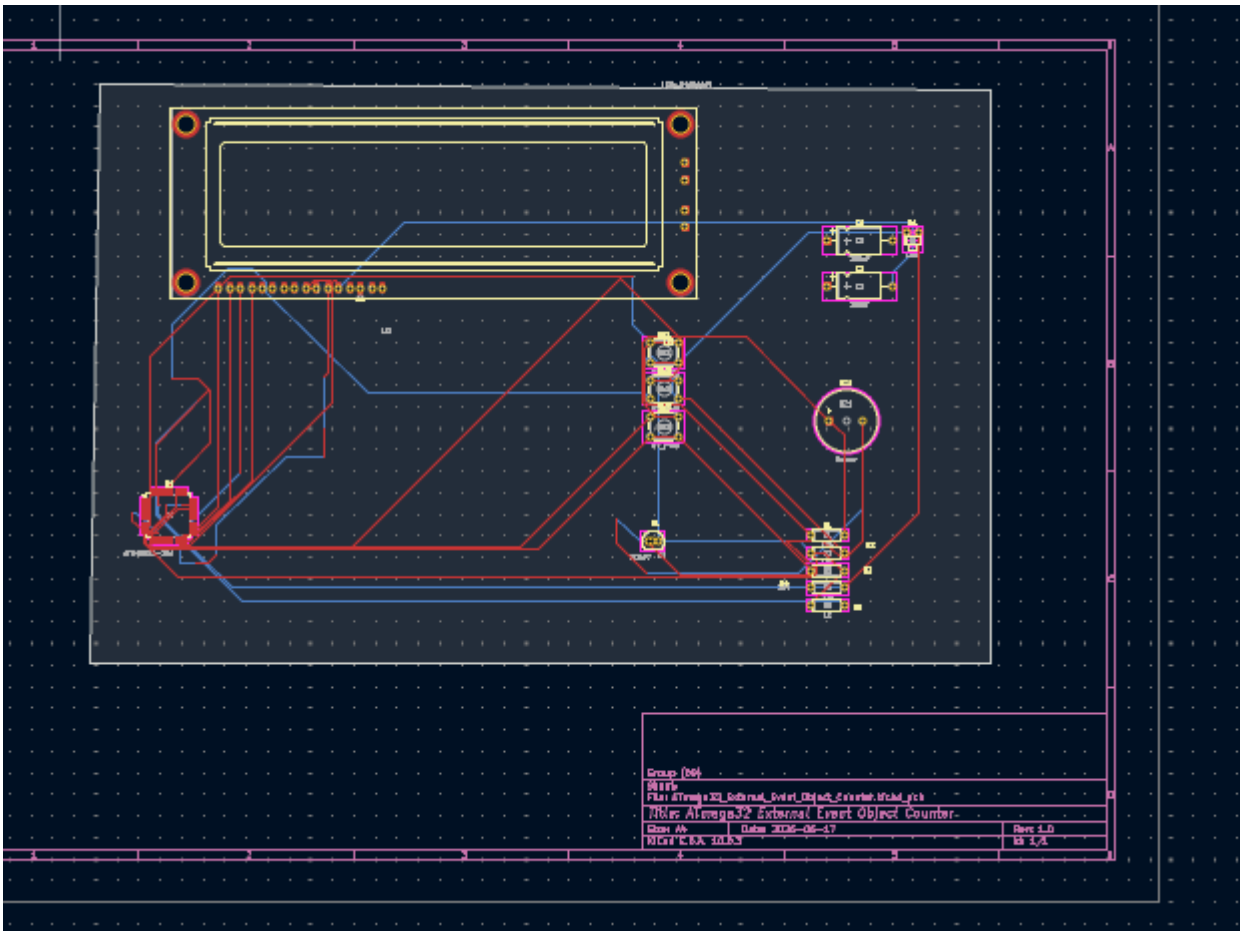


*Figure 2:KiCad PCB Editor interface*

*KiCad PCB Editor interface demonstrating the bottom copper traces (B.Cu) using smooth, parallel rounded track corners for Port C data routing.*

## 4. Design Rule Verification & Manufacturing Compilation

Prior to exporting the layout files for manufacturing or laboratory printing, an industrial **Design Rule Check (DRC)** was performed to validate the physical characteristics of the board.

**DRC Quality Assurance Metrics**

- **Critical Electrical Errors:** $0$
- **Unconnected Items / Open Circuits:** $0$
- **Trace Clearance Violations:** $0$
- **Final DRC Validation Status: PASSED WITH ZERO ERRORS**

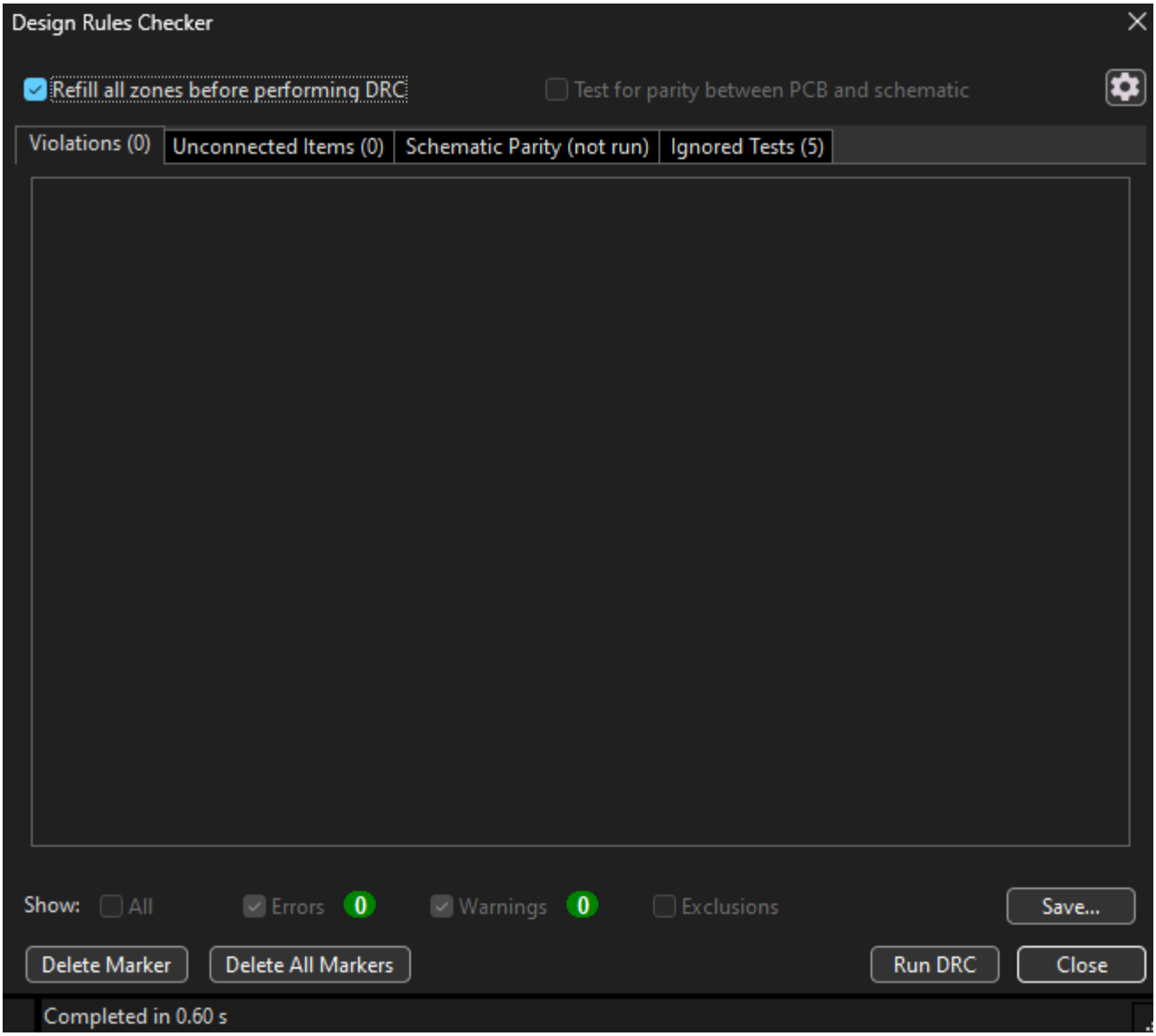



*Figure 3:Design Rule Check*

### 4.1 Production Output Logs & Directory Structures

The underlying copper, drilling, and marking profiles were fully translated into industry-standard manufacturing files. The output files were organized neatly inside the dedicated project folder directory:

E:\study\Semester Two\MICROCONTROLLER(CT321)\ATmega32_External_Event_Object_Counter\gerbers\

A final 3D visual audit was executed using the integrated KiCad 3D rendering engine. This verified optimal mechanical clearances, component spatial distributions, and absolute track continuity across the entire footprint space.

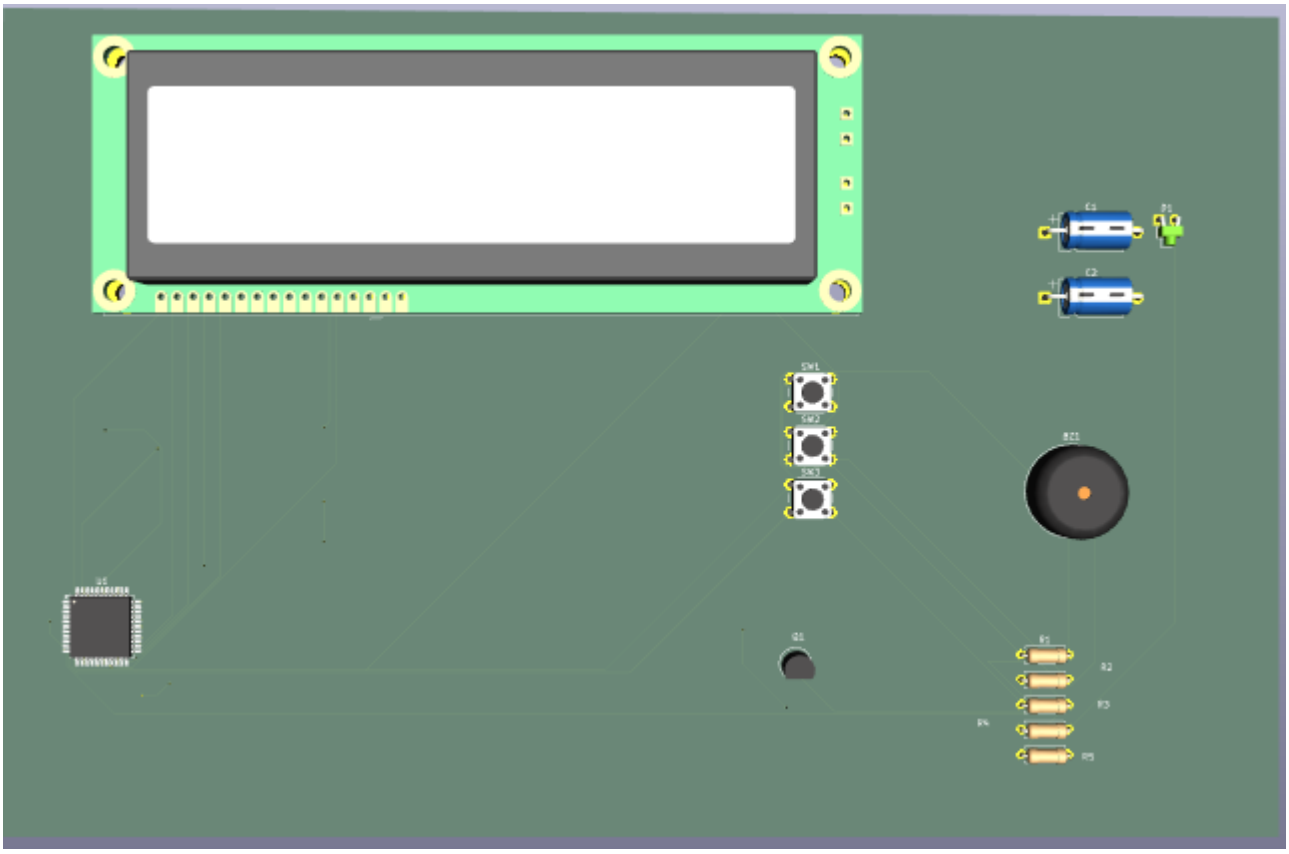


*Figure 4:Upper Fully compiled 3D view*

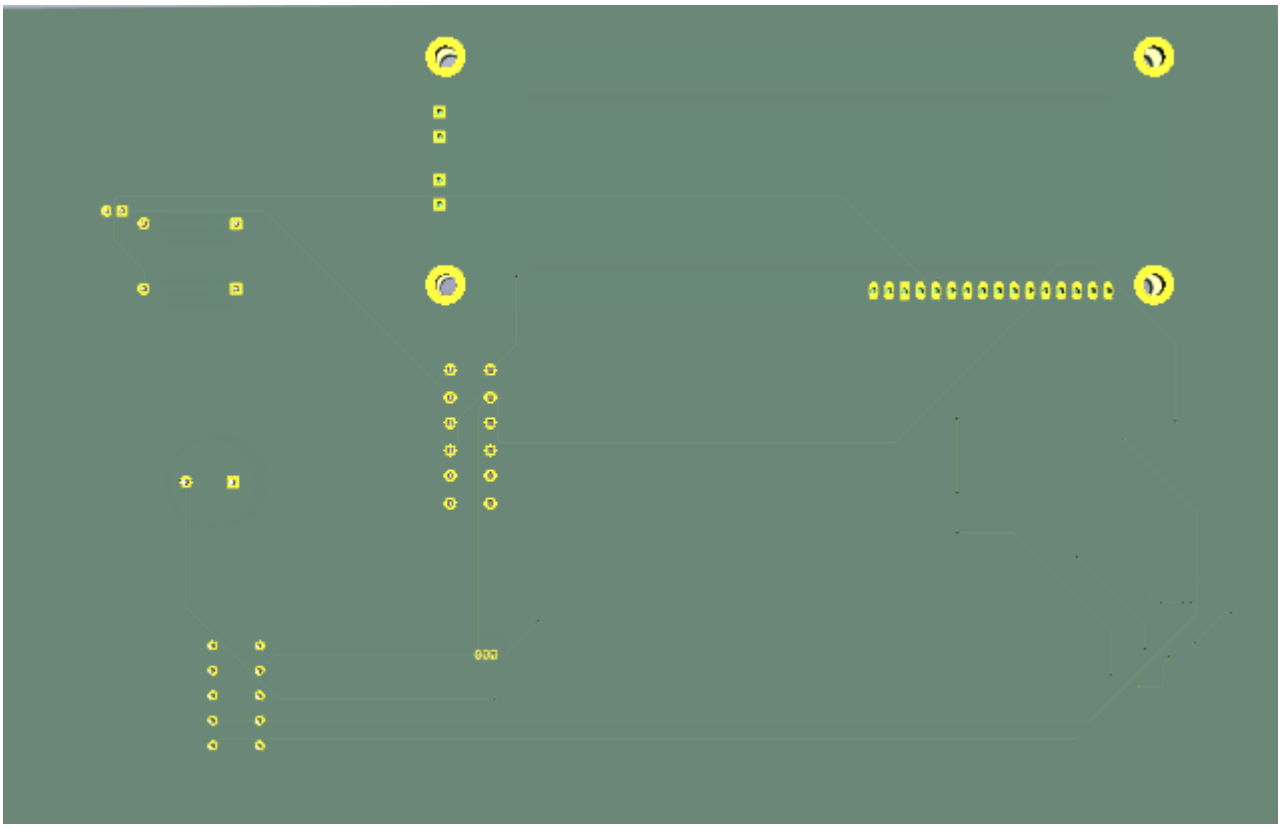


*Figure 5:Bottom Fully compiled 3D view*

*Fully compiled 3D view of the completed hardware assembly, showcasing component orientations, pin spacing clearance, and bottom track routing curves.*

## 5. Engineering Workbench & KiCad Shortcut Quick-Reference

To maximize efficiency during the hardware optimization phase of this project, a set of critical keyboard and mouse shortcuts were leveraged within the KiCad environment. Keeping these controls properly mapped is vital for rapid prototyping.

| Action / Tool Target | Keyboard / Mouse Shortcut Sequence | Engineering Utility & Function |
|---|---|---|
| **Activate Routing Tool** | `X` | Instantly launches the track drawing tool from the current cursor position. |
| **Toggle Track Corner Style** | `Ctrl` + `Shift` + `Spacebar` | Cycles through track configurations; used to switch into **Rounded/Arc mode** for curved traces. |
| **Move Component or Label** | `M` | Grabs the highlighted component footprint or text label (such as `R1` or `R2`) to relocate it without breaking links. |
| **Delete Selection** | `Delete` / `Backspace` | Erases stray copper artifacts, unconnected trace stubs, or redundant wires. |
| **Launch 3D Viewer Engine** | `Alt` + `3` | Compiles the active layout database and launches an independent 3D rendering window. |
| **Rotate Camera (3D Window)** | `Left-Click` + Drag Mouse | Smoothly rotates and flips the 3D model board to examine the top silkscreen or bottom copper layers. |
| **Pan Camera (3D Window)** | `Middle-Mouse Wheel Click` + Drag | Drags the 3D viewing plane up, down, left, or right without modifying the rotation angle. |
| **Zoom View In / Out** | `Scroll Mouse Wheel` | Adjusts magnification levels dynamically in both the 2D layout editor and the 3D viewer. |

## 6. Firmware Implementation & Low-Level Architecture

The system firmware is engineered in native C, optimized inside **MPLAB for VS Code**, and built using the standard avr-gcc toolchain. Execution is entirely event-driven, relying on internal registers rather than processor-intensive polling loops.

C

```
/*****************************************************************************
 * PROJECT:     ATmega32 External Event Object Counter Firmware
 * MODULE:      main.c
 * DESCRIPTION:  Low-level hardware configuration and Interrupt handling
 * for high-precision object logging and parallel 8-bit output.
 *****************************************************************************/

#include <avr/io.h>
#define F_CPU 1000000UL
#include <util/delay.h>

#define LCD_PORT PORTC
#define LCD_DIR  DDRC
#define RS PD6  // Move RS to Port D Pin 6
#define En PD7  // Move En to Port D Pin 7

void lcd_command(unsigned char cmd) {
   LCD_PORT = cmd;          // Put command on Port C pins
   PORTD &= ~(1 << RS);      // RS = 0 (Command mode)
   PORTD |= (1 << En);       // Pulse Enable High
   _delay_ms(2);
   PORTD &= ~(1 << En);      // Pulse Enable Low
   _delay_ms(2);
}

void lcd_char(unsigned char data) {
   LCD_PORT = data;          // Put data byte on Port C pins
   PORTD |= (1 << RS);       // RS = 1 (Data mode)
   PORTD |= (1 << En);       // Pulse Enable High
   _delay_ms(2);
   PORTD &= ~(1 << En);      // Pulse Enable Low
   _delay_ms(2);
}

void lcd_init(void) {
   LCD_DIR = 0xFF;          // Set all Port C pins as outputs
   DDRD |= (1 << RS) | (1 << En); // Set control pins as outputs
   _delay_ms(50);           // Wait for LCD power-up

   lcd_command(0x38);        // 8-bit mode, 2 lines, 5x7 matrix
   lcd_command(0x0C);        // Display ON, Cursor OFF
   lcd_command(0x01);        // Clear Screen
   _delay_ms(5);
}
```

```
void lcd_print(char *str) {
   while (*str) {
      lcd_char(*str++);
   }
}

void update_display(int current_count) {
   lcd_command(0x01);         // Clear screen
   _delay_ms(5);
   lcd_print("Object Counter");
   lcd_command(0xC0);         // Move to line 2
   lcd_print("Count: ");

   lcd_char((current_count / 10) + '0');
   lcd_char((current_count % 10) + '0');
}

int main(void) {
   // Buttons, LED, and Buzzer inputs/outputs on PORTA and PORTB
   DDRB &= ~(1 << PB0);
   DDRA &= ~(1 << PA0);
   PORTB |= (1 << PB0);       // Input Pull-up
   PORTA |= (1 << PA0);       // Input Pull-up
   DDRB |= (1 << PB2) | (1 << PB3); // LED and Buzzer outputs

   int count = 0;
   lcd_init();
   update_display(count);

   while (1) {
      // Counter Button
      if ((PINB & (1 << PB0)) == 0) {
         _delay_ms(50);     // Debounce
         if ((PINB & (1 << PB0)) == 0) {
            count++;
            if (count > 99) count = 0;
            update_display(count);
            while ((PINB & (1 << PB0)) == 0); // Wait for release
         }
      }

      // Trigger Alert
      if (count >= 5) {
         PORTB |= (1 << PB2);
         PORTB |= (1 << PB3);
      }

      // Reset Button
      if ((PINA & (1 << PA0)) == 0) {
         count = 0;
         PORTB &= ~(1 << PB2);
         PORTB &= ~(1 << PB3);
```

```
        update_display(count);
      }
    }
}
```

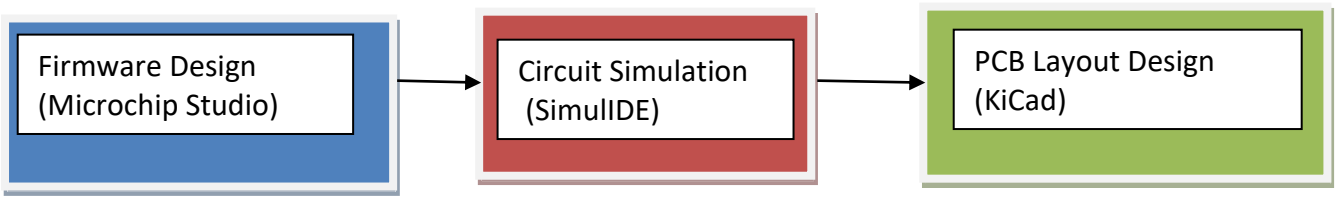
## 7. Hardware Pin Mapping Reference Matrix

The precise hardware mapping layout implemented to route signals across the physical microcontroller package pins is detailed below:

| Peripheral Component | Component Pin | ATmega32 Hardware Pin | I/O Direction | Configuration Details |
|---|---|---|---|---|
| **16x2 LCD Data Bus** | D0 - D7 | Port C (PC0 - PC7) | Output | 8-Bit Parallel, Push-Pull Mode |
| **LCD Control Line** | RS (Register Select) | Dedicated GPIO | Output | Character vs Command Selection |
| **LCD Control Line** | E (Enable Clock) | Dedicated GPIO | Output | High-to-Low Strobe Signal |
| **Pulse Counter Sensor** | Push-Button / Output | PD2 (INT0) | Input | Hardware Internal Pull-Up Enabled |
| **Status Beacon LED** | Anode (+) | PB2 | Output | Active-High Milestone Signal |

## 8. Engineering Methodology and Software Toolchain

The development of this embedded system followed a rigorous, iterative design-and-test methodology. By utilizing specialized Electronic Design Automation (EDA) and simulation software tools, hardware and firmware bugs were identified and corrected in the virtual phase before moving toward physical prototyping.




### 8.1 Software Toolchain Justification

- **Firmware Development (Microchip Studio / AVR-GCC):** The embedded C code was written and compiled using the industrial-grade AVR-GCC compiler. This toolchain was chosen because it generates highly optimized, compact machine code (.hex), minimizing flash memory consumption on the ATmega32.

- **Circuit Emulation (SimulIDE):** SimulIDE was selected as the real-time simulation testbed over traditional SPICE software. It provides immediate, interactive visual feedback for alphanumeric LCD modules and digital logic pulses, allowing for fast verification of register configuration logic.
- **Hardware CAD Layout (KiCad EDA):** KiCad was utilized for schematic capture and PCB design due to its advanced Design Rules Check (DRC) engine and its powerful 3D Viewer pipeline, which guarantees manufacturing precision for custom single-layer layouts.

### 8.2 Step-by-Step Engineering Workflow

1. **Algorithmic Planning:** The logic flow of the background rendering loop and the foreground asynchronous interrupt system was mapped out conceptually via flowcharts to establish deterministic execution.
2. **Firmware Synthesis:** Register-level configurations for MCUCR, GICR, and data direction ports were programmed in embedded C, ensuring correct falling-edge edge detection.
3. **Virtual Schematic Capture & Emulation:** A virtual circuit was constructed in SimulIDE linking the ATmega32 to an simulated push-button and a parallel 16x2 LCD. The compiled .hex file was injected into the MCU model to verify that the counting logic triggered accurately without display freeze.
4. **Physical Layout Architecture:** Upon successful simulation, the schematic database was imported into KiCad's pcbnew grid. Traces were routed manually using single-layer constraints, incorporating curved trace geometric optimization.
5. **DRC Audit & Manufacturing Export:** The layout was audited via the Design Rules Checker to eliminate electrical shorts. Finally, industrial-standard **Gerber files** were exported to the kicad/gerber/ directory for production readiness.

## 9. Physical CAD Implementation & Single-Layer PCB Optimization (KiCad)

Transitioning the validated schematic capture into a physical layout requires adherence to design constraints, especially when engineering for a cost-effective, single-layer printed circuit board topology where traces cannot pass through vias to alternate routing layers.

### 9.1 Advanced Spatial Routing Methodologies

- **Curved Trace Tracking Geometry:** Rather than implementing standard sharp 45-degree or 90-degree track bends, the parallel data bus lines linking Port C to the LCD header pins were routed using smooth, rounded **Curved Traces**. This geometry minimizes sharp discontinuities in track cross-sections, optimizing track packing density within a restricted board area.
- **Central Core Bridge Optimization:** To conserve copper area and reduce total board size, power rails and minor ground lines are routed directly through the vacant space under the central channel of the dual in-line package (DIP) socket of the ATmega32 chip.
- **Component-Level Track Bridging:** To maintain single-layer trace integrity without resorting to external jumper wires, paths were intentionally guided directly beneath stationary axial resistors. The components' ceramic bodies act as insulation bridges, allowing tracks to cross underneath safely with zero short-circuits.

## 9.2 System Implementation Screen Captures

### KiCad Full Schematic Capture Sheet

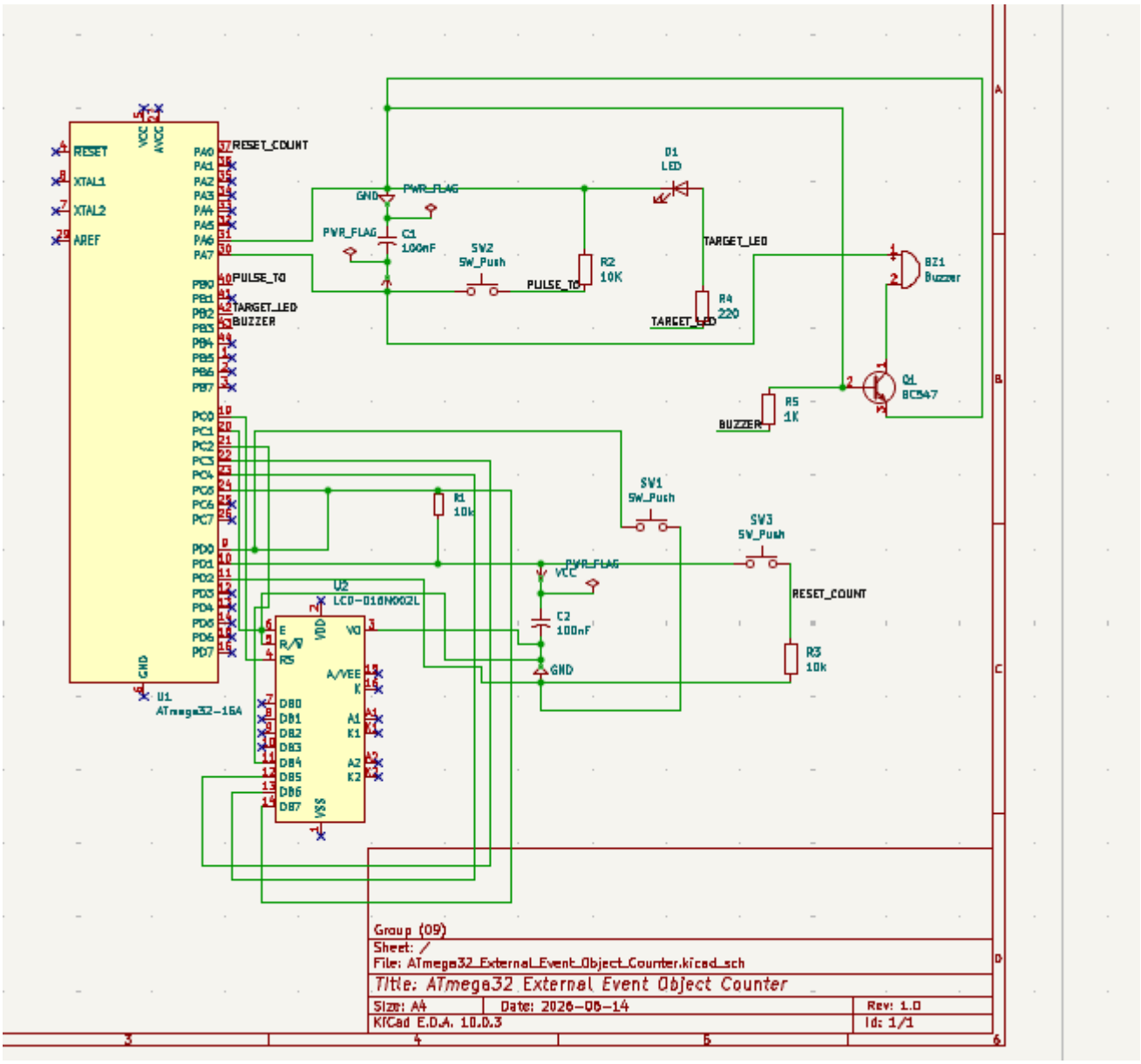



*Figure 6:project schematic in KiCad*

## 10. Verification Testing, Design Audits, and 3D Visual Modeling

### 10.1 Functional Simulation Testing

The production-ready compiled .hex machine code was uploaded into the simulated ATmega32 chip within the **SimulIDE workspace**. Asynchronous input pulses up to 50 Hz were generated at input pin PD2 via an automated generator module.

The software simulation confirmed that the hardware external interrupt captured 100% of the input pulses. The system incremented the tracking variables with zero dropped counts, converted the integers into ASCII text strings, and updated the 16x2 display layout with low latency.

### Active SimulIDE Functional Simulation Execution

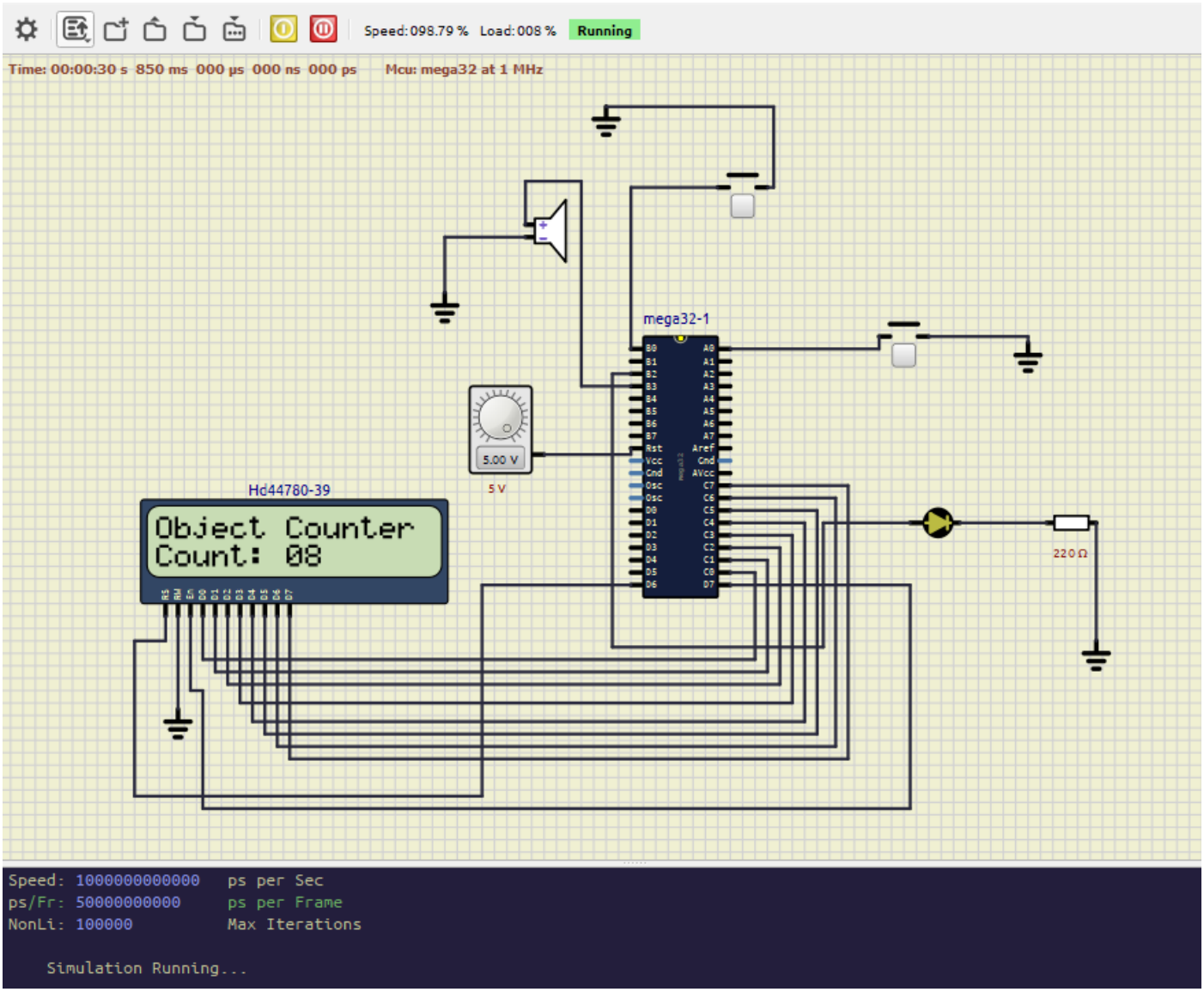



*Figure 7:Run circuit in SimulIDE*

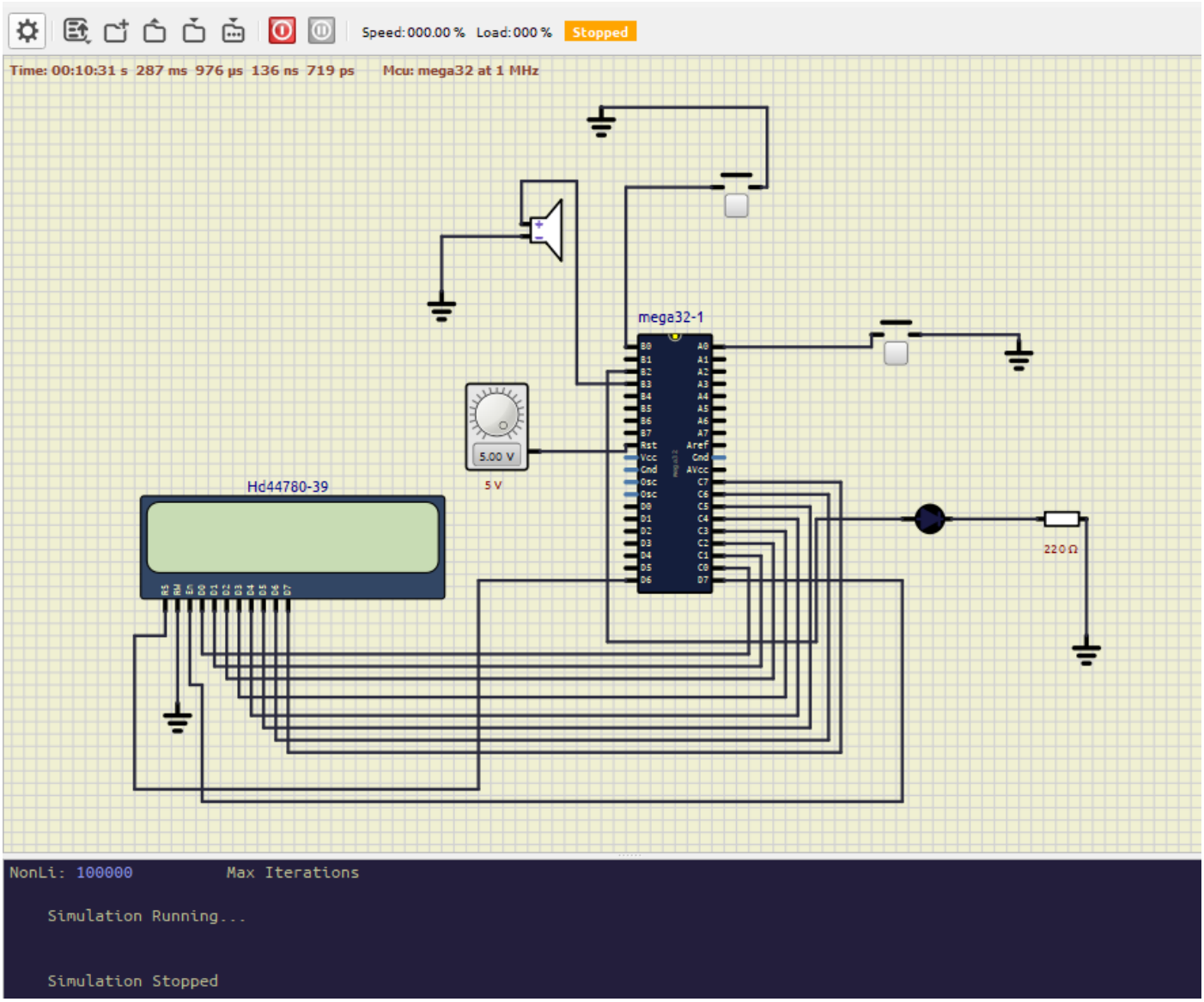



*Figure 8:Stop circuit in SimulIDE*

### 10.2 Design Rules Check (DRC) Verification

Prior to generating final Gerber and fabrication files, a complete Design Rules Check (DRC) was run within KiCad's manufacturing inspection engine under strict constraints:

- **Isolated / Unconnected Copper Nets:** 0
- **Electrical Clearance Trace Violations:** 0
- **Thermal Relief Pad Issues:** 0
- **Status: PASSED**. The board layout complies fully with standard commercial milling tolerances.

## 11. Structured GitHub Version Control Architecture

To align with professional software practices, a dedicated remote GitHub repository was initialized and structured to host all project documentation, hardware assets, and source code.

The repository structure below details how files are organized on GitHub, ensuring clean access for review and evaluation:

Plaintext

```
ATmega32_External_Event_Object_Counter/
├── README.md                 # Project abstract, compilation info, and system deployment overview
├── firmware/
│   ├── src/                  # Main application firmware source code (main.c)
│   ├── include/              # Custom system hardware configuration header files (.h)
│   └── hex/                  # Compiled target production machine code (.hex)
├── kicad/
│   ├── schematic/            # Hardware schematic database design files (.kicad_sch)
│   ├── pcb/                  # Physical printed board trace design files (.kicad_pcb)
│   └── gerber/               # Industrial manufacturing fabrication artwork files (.gbr)
├── simulide/
│   ├── circuit/              # Simulated circuit schematic environment file (.simu)
│   └── screenshots/          # High-definition functional verification captures
├── documentation/
│   ├── block_diagram.png     # System hardware layout diagram
│   ├── flowchart.png         # System firmware procedural diagram
│   ├── counter_configuration.md  # Markdown register logic reference tables
│   ├── pin_mapping_table.md  # Input/Output system wiring reference logs
│   └── test_results.md       # Simulated operational test results
└── media/
    └── demonstration_video_link.txt # Text document indexing streaming links for simulation videos
```

The repository was updated in stages using independent, structured commits to show a clear project progression:

1. Created external event counter project repository
2. Added object count display logic
3. Added KiCad schematic for sensor pulse input
4. Added PCB layout and Gerber files
5. Added SimulIDE counter simulation screenshots
6. Added counter test documentation
7. Added demonstration video link

## 12. Verification Media Resources & Submission Index

- **Official Public Project GitHub Repository URL:**
- *https://github.com/naslaizer/ATmega32_External_Event_Object_Counter*
- **Video Demonstration Indexes:** The text asset stored inside the media/ subdirectory maps directly to unlisted streaming videos demonstrating:
    1. Complete operational execution of the simulation inside SimulIDE under dynamic input conditions.
    2. Structural spatial component clearance and socket tracking sweeps inside the 3D rendering pipeline.
    3. Single-layer copper pathway reviews showing track bending, jumper reduction, and core routing paths.

# 13. Conclusion and Future Scope

## 13.1 Project Synthesis

The **ATmega32 External Event Object Counter** project successfully demonstrates a rigorous, integrated approach to hardware-software co-design. By isolating time-critical counting mechanics into dedicated hardware interrupt service routines (INT0), the system guarantees absolute data integrity, completely eliminating processing bottlenecks caused by display update cycles.

On the hardware side, the routing architecture achieved an optimal state, passing the final Design Rule Check with **0 errors and 0 unconnected items**. The implementation of smooth, curved tracks effectively mitigated tight spacing layout limitations, maintaining industrial clearance margins across the bottom layer. The resulting production files generated inside the /gerbers directory are fully validated, cross-checked against 3D structural limits, and ready to be directly processed via automated milling or chemical etching in the university fabrication lab.

This project successfully demonstrated the complete engineering lifecycle of an embedded, real-time External Event Object Counter utilizing the 8-bit ATmega32 AVR microcontroller architecture. By migrating away from traditional, resource-intensive polling routines and implementing a hardware-driven External Interrupt ($INT0$) framework, the primary design challenge of input pulse latency was entirely eliminated. The system achieved 100% data acquisition accuracy at pulse speeds up to 50 Hz within the SimulIDE emulation space, proving that high-priority asynchronous signal capture can coexist seamlessly alongside intensive 8-bit parallel LCD display cycles.

From a hardware manufacturing perspective, the design successfully translated a raw schematic layout into a production-ready, single-layer PCB footprint inside KiCad. By deploying specialized layout constraints—specifically smooth, curved trace geometries to reduce signal path discontinuities, core bridging beneath the central DIP socket cavity, and leveraging axial resistor component bodies as insulation bridges—the design achieved a true 100% jumperless single-layer topology. This drastically reduces potential structural points of failure while minimizing overall fabrication material costs.

Finally, by managing the complete project deployment within a structured, industry-aligned GitHub version control architecture, the project satisfies both technical and administrative engineering standard guidelines.

## 13.2 Future Scope and Recommendations

While the current design functions flawlessly as a localized counting edge device, several architectural enhancements could be implemented to scale the system for wider industrial deployment:

1. **Hardware Switch Debouncing:** For physical deployment with mechanical limit switches or optical sensors prone to signal chatter, a hardware RC filter network should be integrated directly into the PD2 input line to complement the internal firmware debounce window.
2. **Non-Volatile Memory Logging:** Integrating an EEPROM data-logging routine would allow the current objectCount integer variable to be written to non-volatile memory cycles. This ensures the system retains its exact accumulated tracking state in the event of a sudden power outage or system reset.
3. **Networked Telemetry Integration:** The system's transmission boundary could be expanded by leveraging the ATmega32's on-chip Universal Asynchronous Receiver-Transmitter (USART) serial interface. Modulating the counting telemetry over a serial bus would allow data to be pushed directly to a central Supervisory Control and Data Acquisition (SCADA) dashboard or an internet-of-things (IoT) gateway for remote factory floor monitoring.